

Design and Implementation of a TPS5430-Based DC/DC Buck Converter Power Board for Mobile Robot Applications

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Abstract— This paper presents the design and implementation workflow of a low-voltage DC power board for embedded and mobile robot applications. In the experimental setup, a 220 VAC mains source is first converted to 12 VDC by an external switching power supply, and the designed board then generates regulated 5 V and 3.3 V outputs for logic and control circuits. The 5V rail is produced by a buck converter based on the TPS5430, while the 3.3 V rail is derived through an LDO for MCU and auxiliary logic loads. The proposed design includes reverse-polarity protection, a resettable input fuse, power-good indication, an enable switch, robust input/output connectors, and dedicated test points for debugging and validation. This report summarizes the system architecture, component selection, key design calculations, PCB layout considerations, and prototype verification results. Replace the bracketed items in this draft with the actual component values, fabricated-board photographs, oscilloscope captures, and measured results before final submission.

Keywords—buck converter, TPS5430, robot power board, LDO thermal design, PCB layout, power electronics

I. INTRODUCTION AND DESIGN REQUIREMENTS

The objective of this project is to design a compact and reliable low-voltage power board for a small vehicle or robot platform. In the implemented prototype, the system is powered from a 220 VAC mains source through an external switching power supply, which converts the mains voltage into 12 VDC. The designed onboard power board then generates two regulated output rails: a 5 V main rail and a 3.3 V auxiliary rail. The 5 V output is produced by a TPS5430 [1]-based buck converter to supply logic, sensors, and embedded loads, while the 3.3 V output is derived from an LDO to power the microcontroller and other low-voltage digital circuits. In addition to voltage conversion, the board includes reverse-polarity protection, a resettable input fuse, power-good indicators, a dedicated enable function, robust input/output connectors, and accessible test points for debugging and validation.

Compared with a laboratory bench supply, a robot power board must tolerate vibration, wiring mistakes, and load disturbances caused by control electronics and peripheral devices. Therefore, the design should not only satisfy the

required output-voltage levels, but also address conversion efficiency, thermal performance, EMI-aware PCB routing, mechanical robustness, and safe measurement access. The IEEE report is organized according to these practical design requirements, including system architecture, component selection and calculations, schematic and PCB implementation, prototype verification, and possible future improvements.

The objective of this project is to design a compact and reliable 12 VDC power board for embedded and robotic applications. The designed board accepts a 12 VDC input and provides two regulated output rails, 5 V and 3.3 V, for logic, control, and low-voltage electronic circuits. The 5 V rail is generated by a TPS5430-based buck converter, while the 3.3 V rail is produced through an LDO. In addition to voltage conversion, the board includes reverse-polarity protection, a resettable fuse, power indication, an enable switch, robust connectors, and dedicated test points for debugging and validation.

TABLE I. DESIGN REQUIREMENTS SUMMARY

Item	Target	Design note
Input range	220 VAC mains via external AC/DC switching power supply	The designed board does not process 220 VAC directly; it receives 12 VDC from an external power supply.
Board input voltage	12 VDC	Input is provided by a commercial switching power supply
Main output V_p	5 V	Selectable by feedback network, jumper, or potentiometer.
Auxiliary output	3.3 V via LDO	Used for MCU/logic; thermal limit depends on selected V_p .
Protection	Reverse polarity + resettable fuse	Input protection must prevent damage during field wiring mistakes.
Monitoring	Power-good LED for V_p and	LED threshold should indicate that each rail is valid.

Item	Target	Design note
Control	Enable switch	EN pin requires a defined default state with pull-up or pull-down.
Implementation	Schematic, PCB, prototype, test	Board should expose safe test points for V_{in} , V_p , V_{AUX} , EN, FB, and GND.

II. SYSTEM DESIGN

The overall power path is 12 V input → power switch → reverse-polarity protection → resettable fuse → input filter and bulk capacitors → TPS5430 buck converter → 5 V rail → AMS1117 LDO stage → 3.3 V rail [2]. The enable switch controls the converter through the EN pin, while indicator LEDs and output connectors are provided for both the 5 V and 3.3 V rails. Test points are allocated to V_{in} , GND, 5V, 3V3, EN, and FB so that voltage, ripple, and dynamic behavior can be measured safely during debugging and validation.

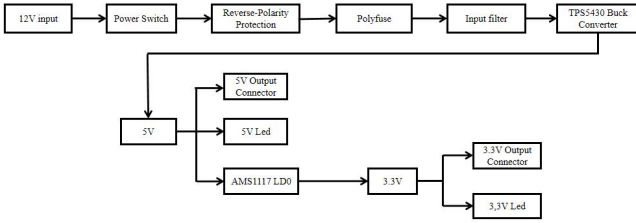


Fig. 1. Block diagram of the proposed 12 V to 5 V/3.3 V power board, including protection, buck conversion, LDO regulation, output indicators, connectors, and test points.

In the proposed architecture, the 3.3 V rail is generated from the regulated 5 V rail through an AMS1117 LDO. This arrangement reduces the voltage drop across the LDO and helps limit thermal dissipation compared with a direct 12 V-to-3.3 V conversion. Therefore, the system is suitable for embedded applications requiring both 5 V and 3.3 V outputs from a single 12 V input source.

III. COMPONENT SELECTION AND CALCULATIONS

A. Reverse-polarity protection and resettable fuse

In this design, a P-channel MOSFET (AO4407A) [4] is employed for reverse-polarity protection instead of a Schottky diode to reduce power loss. When correctly biased, the MOSFET operates with very low $R_{ds(on)}$, resulting in minimal voltage drop and improved efficiency. The maximum input current is estimated as:

$$I_{in,max} \approx (5V \times 3A)/(0.88 \times 12V) \approx 1.42A$$

Compared to a diode with a forward voltage of 0.4V, which would dissipate approximately 0.57W, the MOSFET significantly reduces conduction losses.

A resettable fuse (polyfuse) is placed at the input. It is selected such that $I_{hold} > 1.5A$ and $I_{trip} \approx 2.5\text{--}3A$ to ensure protection under fault conditions while avoiding nuisance tripping.

B. Input capacitor network and feedback divider

The input filtering network consists of bulk and ceramic capacitors, including 4.7μF capacitors and a 0.01μF high-frequency bypass capacitor. This combination reduces input ripple and suppresses switching noise.

The output voltage is set by the feedback divider according to:

$$V_{out} = V_{ref} \times (1 + \frac{R_{top}}{R_{bottom}}) \text{ where } V_{ref} = 1.22V$$

With $R_{bottom} = 3.3k\Omega$ and $R_{top} = 10k\Omega$, the output voltage is approximately 5.0V. Precision resistors (1% tolerance) are used to ensure output accuracy.

C. Inductor, diode, and output-capacitor selection

The inductor value is chosen based on a ripple current target of approximately 30% of the maximum load current. For a 3A load, $\Delta I_L \approx 0.9A$.

Using standard design equations, the calculated inductance is approximately 20μH. Therefore, a 22μH inductor is selected for practical implementation and sufficient margin.

A Schottky diode (B340A, 3A, 40V) [5] is used as the freewheeling diode. The voltage rating exceeds the maximum input voltage, and the current rating satisfies the design requirement.

The output capacitor network includes a 220μF bulk capacitor and a 22μF ceramic capacitor to minimize voltage ripple and improve transient response.

D. LDO thermal analysis for $V_{AUX} = 3.3V$

An AMS1117-3.3 LDO [6] is used to generate the auxiliary 3.3V output from the 5V rail. The power dissipation is given by:

$$P_{LDO} = (V_p - 3.3) \times I_{load}$$

For $V_p = 5V$ and $I_{load} = 0.5A$ so $P \approx 0.85W$. For $I_{load} = 1A$ so $P \approx 1.7W$.

This indicates that thermal management is critical, and the load current should be limited. For higher current requirements, a switching regulator is recommended.

E. Power-Good Monitoring using LM358M

A reliable power-good indication circuit is implemented using the LM358M [7] dual operational amplifier configured as a voltage comparator. The device is powered directly from the 12 V input to ensure independent monitoring regardless of the buck converter operating state.

A fixed reference voltage (V_{ref}) of 1.65 V is generated using a resistive voltage divider ($R_8 = 10k\Omega$ and $R_{11} = 10k\Omega$) derived from the 3.3 V rail. The two comparator channels evaluate this reference against scaled versions of the output voltages.

For the 5 V rail, the output voltage is divided by R_5 (10 kΩ) and R_6 (5.6 kΩ). When the scaled voltage exceeds the reference threshold, corresponding to approximately 4.6 V

($\approx 92\%$ of nominal), the comparator output transitions to a high state, activating the indicator LED.

Similarly, for the 3.3 V rail, the voltage is scaled using R7 (10 k Ω) and R10 (12 k Ω). When the output exceeds approximately 3.02 V ($\approx 91\%$ of nominal), the comparator output switches high, driving the corresponding LED indicator.

This open-loop comparator configuration ensures that both output rails reach acceptable voltage levels before enabling system operation, thereby improving overall reliability and fault visibility.

IV. SCHEMATIC AND PCB LAYOUT

After the electrical design is completed, the schematic should clearly identify the protection stage, buck stage, LDO section, LEDs, switch, and all test points. Important design notes such as feedback-resistor tolerance, connector pin naming, current rating, and jumper configuration for output selection should be visible directly on the schematic to avoid assembly or test mistakes. A clean naming convention also makes debugging faster during prototype bring-up.

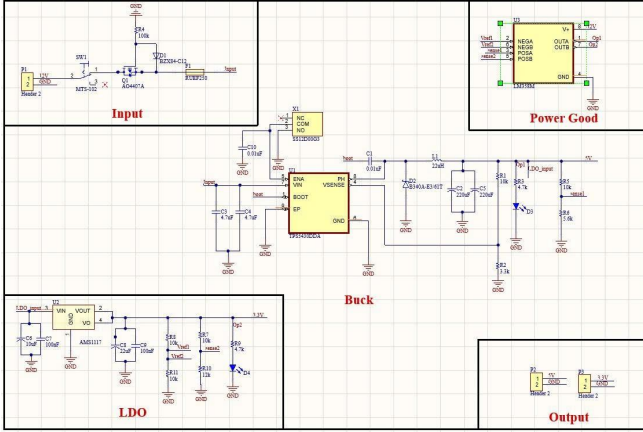


Fig. 2. Schematic overview of the power board with labeled connectors and test points.

The PCB layout is critical for switching-converter performance. The hot loop formed by the input capacitor, switch node, diode, and ground return must be as small as possible. The switch node copper should be minimized and kept away from the feedback divider, the LDO section, and the V_{AUX} trace. Power ground and signal ground should be partitioned sensibly and connected at a quiet point. Thermal copper and multiple vias should be placed under or near the power components, and the V_p path should be wide enough to support 3 A with low resistive loss. The final report should highlight the hot loop directly on the PCB image and explain how the placement follows the datasheet recommendations.

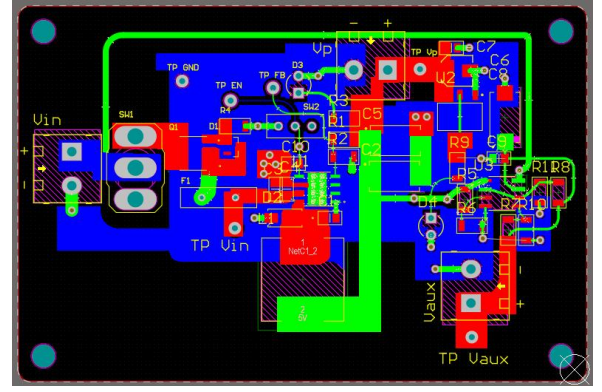


Fig. 3. PCB layout showing the hot loop, switching node, and high-current paths.

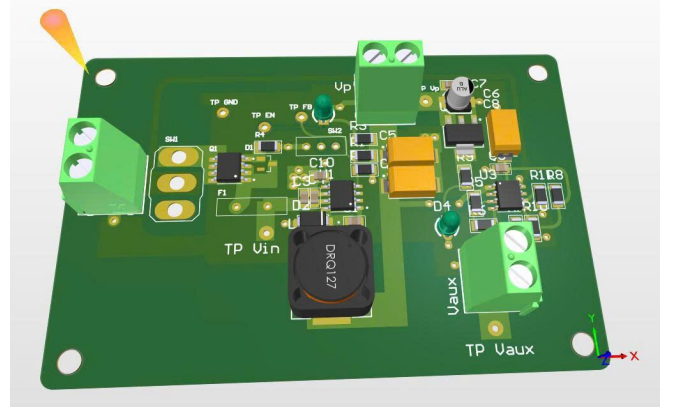


Fig. 4. 3D view of the PCB showing component placement and mechanical arrangement.

V. PROTOTYPE AND TEST RESULTS

Before the first power-up, continuity and short-circuit checks should be performed between V_{in} -GND, V_p -GND, and V_{AUX} -GND. The first energizing step should be carried out with a laboratory supply that includes current limiting. No-load tests should then be recorded at representative input voltages to verify the state of the power-good LEDs and confirm that the control loop and basic wiring are correct before high-power tests are attempted.

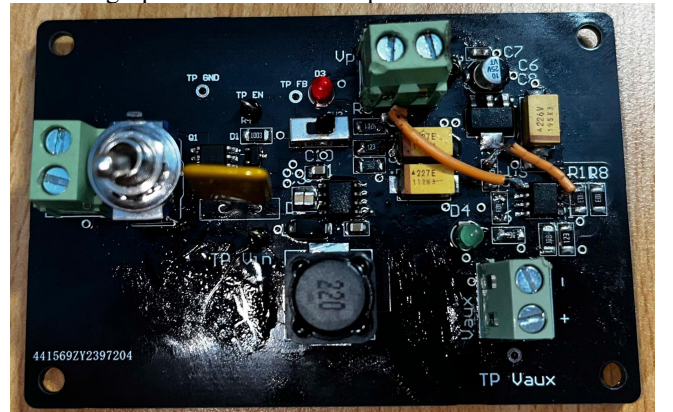


Fig. 5. Prototype after soldering all the components

A. No-Load Testing and Line Drop Compensation

Initial no-load testing at an input of 12V revealed that the default feedback resistor value ($R_{bottom} = 3.3\text{k}\Omega$ and $R_{top} = 10\text{k}\Omega$) yielded exactly 4.92V. To account for voltage drops across wires and connectors when the AGV operates under heavy load, Line Drop Compensation was implemented. The feedback resistor R_{bottom} was replaced

with a $3k6\ \Omega$ resistor and R_{top} was replaced $12k\Omega$. The modified output voltage is verified by:

$$V_{out} = 1.221 \times \left(1 + \frac{12}{3.6}\right) = 5.29V$$

This adjusted starting point ensures that end-point modules receive a stable 5.0V during operation.

B. 5V Main Rail: High-Power Resistive Load and Ripple Measurement

For resistive-load testing, the required dummy load can be calculated from $I = V / R$. At 5V and 3A, the target load is approximately 1.67Ω . To achieve this, a resistor bank consisting of three 5Ω resistors connected in parallel ($R_{eq} = 1.67\Omega$) was used.

During the initial load test, a severe voltage drop to 3.2V was observed. Troubleshooting revealed that the laboratory power supply was insufficient and the test wires had high resistance, confirming the necessity of robust input power and solid connections for the buck converter to operate effectively. Once a proper 12V source was established, the circuit successfully sustained the 3A load at 5.0V.

Oscilloscope captures should document ripple, startup behavior, and load-step response. Using an ATTEN AT7328 analog oscilloscope, the output waveform of the TPS5430 was analyzed under the 3A load. Voltage Ripple: Measured with AC coupling at 20mV/div (1X probe), the peak-to-peak amplitude was 2.5 vertical divisions. The output ripple is $V_{p-p} = 50mV$. This translates to an exceptional 1% ripple noise, indicating a highly effective LC output filter suitable for sensitive logic circuits on the AGV.

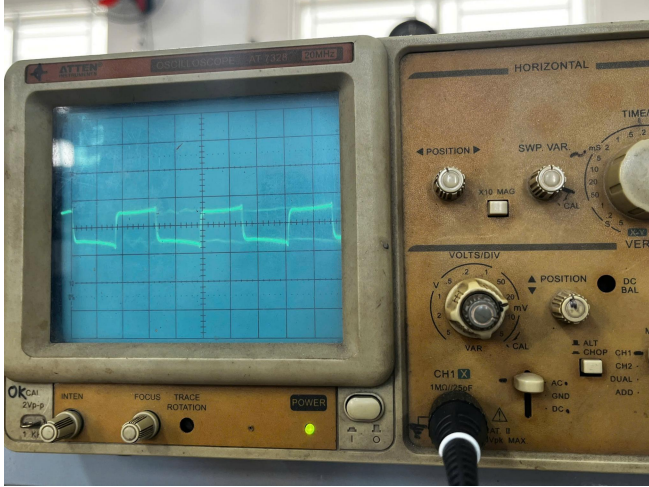


Fig. 6. 5V output ripple under 3A load (50 mVp-p at 20 mV/div, 0.5 us/div).

C. 3.3V Auxiliary Rail: Thermal Limits and Noise Troubleshooting

An AMS1117-3.3 LDO is used to generate the auxiliary 3.3V output from the 5V rail. The power dissipation is given by $P = (V_p - 3.3) \times I$. This indicates that thermal management is critical, and the load current should be limited.

Instead of testing at 1A (which dissipates 1.7W and triggers Thermal Shutdown), a 6 Ohm (5W) ceramic resistor was applied, drawing roughly 0.55A and dissipating a manageable 0.93W on the LDO. Under load, oscilloscope measurements at the 3.3V output revealed an anomalous 1kHz square wave ripple with an amplitude of 24mVp-p.

Because an LDO is a linear device, this periodic square wave was diagnosed not as internal oscillation, but as propagated Hiccup-mode interference from the primary 12V supply dropping out under load, or external ground-loop noise from the laboratory environment.

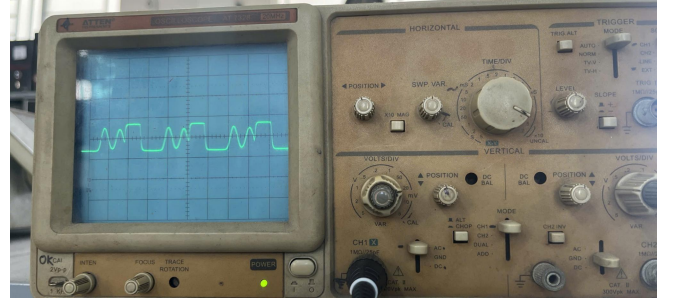


Fig. 7. 3.3V output noise under 0.55A load (24 mVp-p at 20 mV/div, 0.5 us/div).

D. Dynamic Motor Load Testing

To evaluate the transient response and stability of the power board in a real-world AGV application, a dynamic motor load test was conducted using the system's DC gear motor connected to the 5V main rail. If a motor load is used, the report should state the motor type, inrush-current risk, and all safety precautions used during the test. The experimental current profile of the motor was recorded as follows: No-load running current: 0.47A; Startup inrush current: 0.65A ; Stall current (locked rotor): 1.17A.

The TPS5430 buck converter successfully supplied these transient currents. Even during the severe stall condition (1.17A), the 5V rail maintained regulation without triggering overcurrent protection or causing a voltage droop that would reset the logic circuits. As a strict safety precaution, this inductive motor load was completely isolated from the 3.3V auxiliary rail to prevent destructive flyback voltages from damaging the sensitive AMS1117 linear regulator.

TABLE IV. PROTOTYPE MEASUREMENT LOG (REPLACE WITH ACTUAL DATA)

Test	Condition	Expected	Measured	Comment
No-load	$V_{in} = 11.95\ V$	$V_p = 5.29\ V$	5.29V	R_2 modified to $3k\Omega$ for line drop compensation.
No-load	$V_{in} = 11.95\ V$	$V_{AUX} = 3.3\ V$	3.25V	Confirm regulation across the input range.
Resistive load	$V_p = 5\ V, 3\ A$	Ripple low	$V_p = 5.0V$ Ripple = 50mV _{p-p}	Use about 3x5 Ω in parallel 1.67 Ω / >=25 W load.
Resistive load	$V_p = 3.3V, 0.5\ A$	Stable	$V_{AUX} = 3.3V$ Ripple = 24mV _{p-p}	Use 6 Ω /5W load to prevent thermal shutdown.

Test	Condition	Expected	Measured	Comment
Motor load	DC 370 Motor	Handle Inrush/Stall	No load current = 0.47A Inrush current = 0.65A Current limit = 1.17A	Transient response tested during motor stall

VI. CONCLUSION AND FUTURE IMPROVEMENT

This report provides a complete IEEE-style structure for the buck-converter power-board assignment. The design methodology addresses the required voltage range, selectable main output, LDO-derived auxiliary rail, reverse-polarity protection, resettable fuse, power-good indication, enable control, connectors, test points, and PCB implementation constraints. After the prototype is fully tested, the bracketed sections should be replaced by the actual measured regulation, ripple, thermal, and efficiency results so that the report can conclude whether the fabricated board satisfies the assignment targets.

Future work may focus on improved EMI filtering, a synchronous buck stage for higher efficiency, a dedicated switching regulator for the 3.3 V rail, and a more integrated power-good or fault-reporting circuit. These improvements are especially relevant if the board will be used in a dense robot system with long cables, fast motor transients, or tighter thermal constraints.

ACKNOWLEDGMENT

The author would like to thank the course instructor and laboratory staff for their guidance during the design, fabrication, and measurement stages of this project.

REFERENCES

- [1] Texas Instruments, "TPS5430 3-A, Wide Input Range, Step-Down Switching Regulator," datasheet, latest revision. Replace with the exact document code and revision used in the project.
- [2] Texas Instruments, "Layout Guidelines for Switching Power Supplies," application note. Replace with the exact application note referenced during PCB design.
- [3] Texas Instruments, "WEBENCH Power Designer," online design tool, used to cross-check the component values and operating conditions.
- [4] UMW AO4407A P-Channel MOSFET datasheet.
- [5] Vishay General Semiconductor B340A-E3/61T Diode Schottky datasheet.
- [6] Advanced Monolithic Systems AMS1117, 1A LOW DROPOUT VOLTAGE REGULATOR datasheet.
- [7] Texas Instruments, LM358M Industry-Standard Dual Operational Amplifiers datasheet